Proofs

Theorem 1.1. *(The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0$ $\exists!\, q, r \in \mathbb{Z}$ with $0 \le r < b$ s.t. $a = bq + r$.*

Proof. (*Existance*): Let $R = \{a - bm : m \in \mathbb{Z} \;\&\; a - bm \ge 0\}$, then as: if $a \ge 0$, $a - b \cdot 0 = a \ge 0$ so $a \in R$; if $a < 0$, $a - b \cdot a = a(1 - b) \ge 0$ so $a - b \cdot a \in R$, $R \ne \varnothing$. Thus, take $r = \min(R)$ & so $0 \le r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q+1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \le r < b$. (*Uniqueness*): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 \le r', r < b$ & $a = bq + r = bq' + r'$, also WLOG take $r' \le r$. So with $\frac{r-r'}{b} \in (-1, 1)$, & $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. □

Euclidian Algorithm:
Let $a, b \in \mathbb{Z}^+$, apply Th[m] 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1 q_2 + r_2 \\ r_1 &= r_2 q_3 + r_3 \\ &\;\;\vdots \\ r_{n-1} &= r_n q_{n+1}, \end{aligned}$$

Here, r_n is the last non-zero remainder and $\gcd(a, b) = r_n$. To prove this relation we show: $G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where $G(a, b) = \{n \in \mathbb{Z} : n|a \;\&\; n|b\}$.

Reverse Euclid:
Given a completed algorithm we can obtain $x, y \in \mathbb{Z}$ s.t. $xa + yb = \gcd(a, b)$ i.e. taking the start of a Euclidian algorithm:

$$\begin{aligned} a = q_1 b + r_1 \quad &\Rightarrow \quad r_1 = a - q_1 b \\ b = q_2 r_1 + r_2 \quad &\Rightarrow r_2 \;\; = b - q_2 r_1 \\ &\qquad\quad = b - q_2(a - q_1 b) \\ &\qquad\quad = -q_2 a + (1 + q_1 q_2) b \\ r_1 = q_3 r_2 + r_3 \quad &\Rightarrow r_3 \;\; = r_1 - q_3 r_2 \\ &\qquad\quad = (a - q_1 b) \\ &\qquad\qquad - q_3(-q_2 a + (1 + q_1 q_2) b) \\ &\qquad\quad = (1 + q_2 q_3) a \\ &\qquad\qquad - (q_1 + q_3 + q_1 q_2 q_3) b \\ &\;\;\vdots \end{aligned}$$

Continue substituting until only a, b remain:
$$r_n = \gcd(a, b) = xa + yb, \quad x, y \in \mathbb{Z}.$$

Theorem 1.9. *Every integer $n \ge 2$ can be written as a product of primes.*

Proof. (*By strong induction*):
(*Base case*, $n = 2$): Trivially true, $n = 2$ a product of primes.
(*Inductive Hypothesis*): Suppose for $n \in 2, 3, \ldots, k$ $n = p_{n,1} p_{n,2} \cdots p_{n,m_n}$. Then for $n = k + 1$, if $k + 1$ is prime we are done. If not, $k + 1 = ab$ where $a, b \in \{2, 3, \ldots, k\}$, so by the hypothesis a and b are products of primes; hence $k + 1$ is too.
So $\forall n \in \mathbb{N}$ with $n \ge 2$, we are done. □

Theorem 1.10. *(Euclid's property of primes) $\forall p \in$ Primes & $\forall a, b \in \mathbb{Z}$. $p|ab \implies p|a$ or $p|b$.*

Proof. (*Case $p|a$*): We are done.
(*Case $p \nmid a$*): Then using the Euclidian algorithm we obtain:
$1 = na - mp$ with $n, m \in \mathbb{Z}$. So
$p|b \iff p|(na - mp)b \iff p|n(ab) - mbp \impliedby p|ab$. □
the chain of implications uses a proposition in the notes which states:
$p|a \;\&\; p|b \implies p|a + b \;\&\; p|ab$.

Theorem 1.12. *(Fundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$ with $n \ge 2$, $\exists!\, p_1, p_2, \ldots, p_r \in$ Primes up to ordering, s.t.: $n = p_1 p_2 \cdots p_r$.*

Proof. The existance is from Th[m] 1.9. For uniqueness, suppose $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, then cancel common primes, suppose the result is not 1 i.e. we are left with $p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$ with $p_i \ne q_j, \forall i \in \{1, \ldots, n\}$ & $\forall j \in \{1, \ldots, m\}$, after reindexing. Then $p_1 | q_1 q_2 \cdots q_m$ so $\exists j \in \{1, \ldots, m\} p_1 | q_j$, but q_j is prime so $p_1 = q_j$, ↯. □

Lemma 1.16. *$\forall a, b \in R$: $0a = a0 = 0$; $a(-b) = (-a)b = -(ab)$; & $(-a)(-b) = ab$*

Proof. $0a + 0 = 0a = (0 + 0)a = 0a + 0a$ so $0a = 0$.
$ab + a(-b) = a(b - b) = a0 = 0 = (a - a)b = ab + (-a)b$ so by the Groups & Geo[m] we have $-(ab)$ is the unique element s.t. $ab - (ab) = 0$, $a(-b) = -(ab) = (-a)b$.
$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. □

Lemma 1.18. *(Subring Test) Let R be a ring & $S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:*
1. $1_R \in S$,
2. $r + s, rs \in S$, &
3. $-r \in S$

Proof. By 1 & 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ & $\times$, & distributivity of $\times$ over $+$ are all inheritited from R. Note, as they shair operations they shair the identities of the operations, so as, $0_R \ne 1_R$ in R we have $0_S = 0_R \ne 1_R = 1_S$ in S, so (R1)-(R4) are satisfied. □

Lemma 2.2. *Suppose $Char(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$; & (ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.*

Proof. (i):
$$n \cdot r = \underbrace{r + r + \cdots + r}_{n \text{ times}} = r\underbrace{(1 + 1 + \cdots + 1)}_{n \text{ times}} = r0 = 0$$

(ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division Th_m with $m = nq + r$ where $q, r \in \mathbb{Z}$ & $0 \le r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 = q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as otherwise $n > r \in \mathbb{Z}^+$ & $r \cdot 1 = 0$, a ↯ of n's def[n].
(*impliedby*): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$. □

Lemma 2.5. *If S is a subring of R & R is a domain then S is a domain.*

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R, hence $r = 0$ or $s = 0$. □

Proposition 2.7. *Suppose R is a domain. Then $R[X]$ is a domain.*

Proof. We use the contrapositive of the definition of a domain [i.e.:R a domain iff $\forall 0 \ne r, s \in R, rs \ne 0$]. Let $0 \ne f, g \in R[X]$, taking the leading coefficients of f & g as $a \ne 0$ & $b \ne 0$, respectively, the leading coefficent of fg as $ab \ne 0$, hence $fg \ne 0$. □

Lemma 2.11. *If R is a ring & $r \in R$ has both a left & right inverse then these equal & unique.*

Proof. (Equal): Choose, $rs = 1 = tr$, then: $s = 1s = trs = t1 = t$.
(Unique right inverse): Let $rs = 1 = rs'$ then $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is choosen s.t. $tr = 1$.
(Unique left inverse): Let $tr = 1 = t'r$ then $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is choosen s.t. $rs = 1$ □

Proposition 2.13. *Every division ring is a domain (& subsequently every field is an integral domain).*

Lemma 2.12. *The following conditions for an integer $n \ge 2$ are equivalent:*
(i) $\mathbb{Z}_n$ is an integral domain;
(ii) $\mathbb{Z}_n$ is a field;
(iii) n is prime.

Proof. (i) $\implies$ (ii): Suppose $\mathbb{Z}_n$ is a domain & let $[m]_n \in \mathbb{Z}_n$ & $[m]_n \ne 0$. Consider the list $[0]_n[m]_n, [1]_n[m]_n, \ldots, [n-1]_n[m]_n$, each element in the list is distinct as, suppose otherwise then $[i]_n[m]_n = [j]_n[m]_n$ for some distinct $i, j \in \{0, 1, \ldots, n-1\}$, then $([i]_n - [j]_n)[m]_n = 0$ but $\mathbb{Z}_n$ is suppose to be a domain & $[m]_n \ne 0$ so $[i]_n - [j]_n = [0]_n$ i.e. $[i]_n = [j]_n$, ↯. Then as there are n elements of the list each in $\mathbb{Z}_n$, then $\{[0m]_n, [1m]_n, \ldots, [(n-1)m]_n\} = \mathbb{Z}_n$. So some multiple of $[m]_n$ say $[r]_n \in \mathbb{Z}_n$ is s.t. $[m]_n[r]_n = [r]_n[m]_n = [1]_n$.

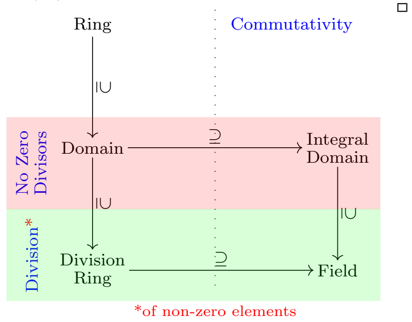
(ii) $\implies$ (iii): Suppose $\mathbb{Z}_n$ is a field then suppose $m, l \in \mathbb{N}$ are s.t. $ml = n$, so $0 < m \le l \le n$. Then $[m]_n[l]_n = [ml]_n = [n]_n = [0]_n$ so as fields are domains, $[m]_n = [0]_n = [n]_n$ or $[l]_n = [0]_n = [n]_n$ & so, by the equivalence class definition, $m = n$ or $l = n$ as neither are 0. So if $m = n$, $l = 1$ & if $l = n$, $m = 1$, so the only factors of n are 1 & n in $\mathbb{N}$, i.e. n is prime.
(iii) $\implies$ (i): Suppose n is prime & let $[m]_n, [l]_n \in \mathbb{Z}_n$ s.t. $[m]_n[l]_n = 0$ i.e. $[ml]_n = [0]$ defintionally $n \mid ml$ so by Th[m] 1.10 $n \mid m$ or $n \mid l$ i.e. $[m]_n = [0]_n$ or $[l]_n = [0]_n$. So with $\mathbb{Z}_n$'s commutativity, $\mathbb{Z}_n$ is an integral domain. □

Proposition T1. *In a domain R if $ab = ac$ or $ba = ca$ for $a, b, c \in R$ & $a \ne 0$ then $b = c$.*

Proof. If $ab = ac$, then $a(b - c) = 0$. If $ba = ca$, then $(b - c)a = 0$. Since $a \ne 0$ and R is a domain, $b - c = 0$. □

Lemma 2.15. *In a ring R, the set of units R^* forms a group under multiplication.*

Proof. (G0): $1 \cdot 1 = 1$ so $1 \in R^*$ i.e. $R^* \ne \varnothing$; (G1): given (R2) we have associativity & given $r, s \in R^*$, $(rs)(s^{-1}r^{-1}) = r(ss^{-1})r^{-1} = 1$ & $(s^{-1}r^{-1})(rs) = s^{-1}(r^{-1}r)s = 1$, so $rs \in R^*$ as required; (G2): (R4); & (G3): Let $r \in R^*$ so r is a unit so $r^{-1} \in R^*$ exists, which itself has inverse $(r^{-1})^{-1} = r \in R$, so $r^{-1} \in R^*$, as required. □



Lemma 3.3. *Suppose $\phi : R \to S$ is an isomorphism. Then: (i) $\phi(1_R) = 1_S$; (ii) $\phi(0_R) = 0_S$; (iii) $\phi(-r) = -\phi(r)$, $\forall r \in R$; (iv) $r \in R$ is invertible iff $\phi(r) \in S$ is invertible and, in that case, $(\phi(r))^{-1} = \phi(r^{-1})$; (v) $r \in R$ is nilpotent iff $\phi(r)$ is nilpotent and, in that case, they have the same index of nilpotence.*

Lemma 3.7. *Suppose $\vartheta : R \to S$ is a homomorphism. Then: (i) $\vartheta(0) = 0$; (ii) $\vartheta(-r) = -\vartheta(r)$, $\forall r \in R$; (iii) if $r \in R$ is invertible, then $\vartheta(r) \in S$ is invertible and $(\vartheta(r))^{-1} = \vartheta(r^{-1})$; (iv) if $r \in R$ is nilpotent, then $\vartheta(r)$ is nilpotent with index $\le$ the index of r; (v) $\mathrm{Im}(\vartheta)$ is a subring of S.*

Lemma 3.10. *(i) If $\vartheta : R \to S$ & $\beta : S \to T$ are homomorphisms then $\beta \circ \vartheta : R \to T$ is a homomorphism. (ii) If ϑ, β are embeddings then $\beta \circ \vartheta$ is an embedding. (iii) If ϑ, β are homomorphisms and $\beta \circ \vartheta$ is an embedding then ϑ is an embedding.*

Lemma 3.15. *If $\vartheta : R \to S$ is a homomorphism then ϑ is injective iff $\ker(\vartheta) = 0$.*

Lemma 3.17. *(i) Suppose $\vartheta : R \to S$ is a homomorphism. Then $\ker(\vartheta)$ is a subgroup of $(R, +)$. (ii) Let $r, r' \in R$. Then $\vartheta(r) = \vartheta(r')$ iff $r - r' \in \ker(\vartheta)$ iff r & r' belong to the same coset of $\ker(\vartheta)$ in R.*

Proposition 3.22. *A commutative ring R is a field iff the only ideals of R are R and $\{0\}$.*

Proof. ($\implies$) Suppose R is a field, then clearly $\{0\} \lhd R$ now suppose $\{0\} \ne I \lhd R$ so then by defintion I contains a non-zero element of R say $r \in I$. Now let $s \in R$, so, as R is a field r^{-1} exists, then $s = (sr^{-1})r \in I$, hence $I = R$, as requried. ($\impliedby$) Suppose the only ideals of R are R and $\{0\}$. Now let $r \in R \setminus \{0\}$ then $\langle r \rangle = R$ hence $1 \in \langle r \rangle$ so by the defintion of the principle ideal in a commutative ring $1 = sr (= rs)$ for some $s \in R$, hence s is r's inverse all non-zero elements in R have inverses, hence R is a field. □

Proposition 3.24. *If $\vartheta : R \to S$ is a homomorphism of rings, then $\ker(\vartheta)$ is an ideal of R.*

Proof. We show $\ker(\vartheta)$ satisfies the ideal conditions. Firstly $0 \in \ker(\vartheta)$. Let $r, r' \in \ker(\vartheta)$. Then $\vartheta(r + r') = \vartheta(r) + \vartheta(r') = 0 + 0 = 0$, so $r + r' \in \ker(\vartheta)$, and $\vartheta(-r) = -\vartheta(r) = 0$, so $-r \in \ker(\vartheta)$. Let $s \in R$. Then $\vartheta(sr) = \vartheta(s)\vartheta(r) = \vartheta(s)0 = 0$ and similarly $\vartheta(rs) = 0$. Therefore $sr, rs \in \ker(\vartheta)$. □

Corollary 3.25. *If $\vartheta : R \to S$ is a homomorphism of rings and R is a field then ϑ is a monomorphism.*

Proof. By Pro[p] 3.24, $\ker(\vartheta) \lhd R$, and so Pro[p] 3.22 implies $\ker(\vartheta) = \{0\}$ or R. We cannot have $\ker(\vartheta) = R$, as then $1 = \vartheta(1) = 0$. Therefore $\ker(\vartheta) = \{0\}$ and ϑ is a monomorphism. □

Proposition 3.26. *Suppose I and J are ideals of the ring R. Then: (i) $I + J = \{a + b : a \in I, b \in J\}$ is an ideal; (ii) $I \cap J$ is an ideal; (iii) if $\{I_\lambda\}_\lambda$ is any collection of ideals of R, then $\bigcap_\lambda I_\lambda$ is an ideal.*

Lemma 4.2. *(i) The operations $+$ and $\times$ on R/I are well-defined, i.e. if $r + I = r' + I$ and $s + I = s' + I$, then $(r + s) + I = (r' + s') + I$ and $rs + I = r's' + I$. (ii) With these operations, R/I forms a ring. The zero element is $0 + I (= I)$, the identity is $1 + I$, and $-(r + I) = (-r) + I$. If r^{-1} exists in R, then $r + I$ is invertible in R/I with inverse $r^{-1} + I$.*

Proof. For well-definedness of $+$ and $\times$, suppose $r + I = r' + I$ and $s + I = s' + I$. Then $r - r' \in I$ and $s - s' \in I$. Hence
$$(r + s) - (r' + s') = (r - r') + (s - s') \in I,$$
so $(r + s) + I = (r' + s') + I$. Also, since $I \lhd R$,
$$rs - r's' = (r - r')s + r'(s - s') \in I,$$
hence $rs + I = r's' + I$. Thus both operations are independent of the chosen representatives. □

Lemma 4.4. *Let I be a proper ideal of R. (i) The map $\pi : R \to R/I$, $\pi(r) = r + I$, is a surjective ring homomorphism with kernel I. (ii) Let $\vartheta : R \to S$ be a homomorphism with $I \subseteq \ker(\vartheta)$. There is an induced homomorphism $\vartheta' : R/I \to S$ with $\vartheta' \circ \pi = \vartheta$.*

Proof. (i): Every element of R/I has form $r + I = \pi(r)$, so π is surjective. Also $\pi(r + s) = (r + s) + I = (r + I) + (s + I) = \pi(r) + \pi(s)$, similarly $\pi(rs) = \pi(r)\pi(s)$, and $\pi(1) = 1 + I$. Hence π is a homomorphism. Moreover
$$\ker(\pi) = \{r \in R : \pi(r) = 0 + I\} = \{r \in R : r + I = I\} = I.$$
(ii): Define $\vartheta' : R/I \to S$ by $\vartheta'(r + I) = \vartheta(r)$. If $r + I = r' + I$, then $r - r' \in I \subseteq \ker(\vartheta)$, so $\vartheta(r - r') = 0$, hence $\vartheta(r) = \vartheta(r')$. Thus ϑ' is well-defined. Also
$\vartheta'((r + I) + (s + I)) = \vartheta'(r + s + I) = \vartheta(r + s) = \vartheta(r) + \vartheta(s)$, and similarly for multiplication, while $\vartheta'(1 + I) = 1$. Thus ϑ' is a homomorphism and $\vartheta' \circ \pi = \vartheta$. □

Theorem 4.5. *(Fundamental Isomorphism Theorem) Let $\vartheta : R \to S$ be a ring homomorphism. Then $R/\ker(\vartheta) \simeq \mathrm{Im}(\vartheta)$.*

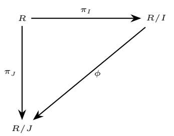
Proof. Let $I = \ker(\vartheta)$. By Lemma 4.4(ii), $\vartheta' : R/I \to S$ defined by $\vartheta'(r + I) = \vartheta(r)$ is a homomorphism. We have $\vartheta'(r + I) = 0$ iff $\vartheta(r) = 0$ iff $r \in \ker(\vartheta) = I$ iff $r + I = 0 + I$, so $\ker(\vartheta') = \{0 + I\}$. By Lemma 3.15, ϑ' is injective. Also $\mathrm{Im}(\vartheta') = \mathrm{Im}(\vartheta)$. Therefore $R/\ker(\vartheta) \simeq \mathrm{Im}(\vartheta)$. □

Theorem 4.8. *Let I be a proper ideal of R. Then ideals $J \lhd R$ with $J \ge I$ are in one-to-one correspondence with ideals $K \lhd R/I$, via $J \mapsto \pi J = \{\pi(r) : r \in J\}$ and $K \mapsto \pi^{-1}K = \{r \in R : \pi(r) \in K\}$.*

Proof. Let $A = \{J \lhd R : J \ge I\}$ and $B = \{K \lhd R/I\}$. Define $f : A \to B$ by $f(J) = \pi J$ and $g : B \to A$ by $g(K) = \pi^{-1}K$. First, to check these maps are well-defined, we check their outputs really lie in the stated codomains. If $J \in A$, then $\pi J \lhd R/I$, so $f(J) \in B$: $0 = \pi(0) \in \pi J$; if $b = \pi(a), b' = \pi(a') \in \pi J$, then $b + b' = \pi(a + a') \in \pi J$; if $s = \pi(r) \in R/I$, then $bs = \pi(ar) \in \pi J$ and similarly $sb \in \pi J$. If $K \in B$, then $\pi^{-1}K \lhd R$ and contains I, so $g(K) \in A$: $0 \in \pi^{-1}K$; if $a, a' \in \pi^{-1}K$, then $\pi(a + a') = \pi(a) + \pi(a') \in K$; and if $r \in R$, then $\pi(ar) = \pi(a)\pi(r) \in K$ and similarly $\pi(ra) \in K$. Also $I \subseteq \pi^{-1}K$, since $a \in I \implies \pi(a) = 0 \in K$.
Now g is the inverse of f. For $J \in A$, clearly $J \subseteq \pi^{-1}(\pi J)$. Conversely, if $a \in \pi^{-1}(\pi J)$, then $\pi(a) = \pi(a')$ for some $a' \in J$. Thus $a - a' \in \ker(\pi) = I \subseteq J$, so $a = a' + (a - a') \in J$. Hence $\pi^{-1}(\pi J) = J$. For $K \in B$, if $b \in K$, surjectivity of π gives $b = \pi(r)$, and $r \in \pi^{-1}K$, so $b \in \pi(\pi^{-1}K)$. Conversely, if $b \in \pi(\pi^{-1}K)$, then $b = \pi(r)$ for some $r \in \pi^{-1}K$, hence $b \in K$. Thus $\pi(\pi^{-1}K) = K$. Therefore f and g are inverse bijections. □

Theorem 4.10. *If $I \le J$ are proper ideals of R, so $J/I \lhd R/I$, then $(R/I)/(J/I) \simeq R/J$.*

Proof. Use the Fundamental Isomorphism Theorem on $\phi : R/I \to R/J$, $\phi(r + I) = r + J$. This is a surjective homomorphism, and $\ker(\phi) = \{r + I : r \in J\} = J/I$. Hence $(R/I)/(J/I) \simeq R/J$. □



Corollary 4.13. *If R is commutative then a proper ideal $I \lhd R$ is maximal iff the quotient ring R/I is a field.*

Proof. By Pro[p] 3.22, it is enough to show I maximal iff the only ideals of R/I are $\{0\}$ and R/I. If I is maximal and $K \lhd R/I$, then by Th[m] 4.8, $\pi^{-1}K$ is an ideal of R containing I, so $\pi^{-1}K = I$ or R. Thus $K = \{0\}$ or R/I. Conversely, if R/I is a field and $I \le J \lhd R$, then $\pi J \lhd R/I$, so by Pro[p] 3.22, $\pi J = \{0\}$ or R/I. By Th[m] 4.8, this gives $J = I$ or $J = R$. Hence I is maximal. □

Theorem 4.16. *Let I be a proper ideal of a commutative ring R. Then I is prime iff R/I is an integral domain.*

Proof. Since R is commutative, so is R/I, and since I is proper, $0 + I \ne 1 + I$. If I is prime and $(r + I)(s + I) = 0 + I$, then $rs \in I$, so $r \in I$ or $s \in I$, i.e. $r + I = 0 + I$ or $s + I = 0 + I$. Conversely, if R/I is an integral domain and $rs \in I$, then $(r + I)(s + I) = 0 + I$, so $r + I = 0 + I$ or $s + I = 0 + I$, hence $r \in I$ or $s \in I$. Thus I is prime. □

Theorem 5.1. *(Division Theorem for Polynomials) Let K be a field and $f, g \in K[X]$ with $g \ne 0$. Then there are unique $q, r \in K[X]$ with $f = qg + r$ and $\deg(r) < \deg(g)$ or $r = 0$.*

Proof. Let $A = \{f - hg : h \in K[X]\}$. Choose $r = f - qg \in A$ of least degree. If $r = 0$ or $\deg(r) < \deg(g)$, done. Otherwise, since K is a field, a suitable scalar multiple of $X^{\deg r - \deg g} g$ can be subtracted from r to cancel its leading term, giving $r - hg = f - (q + h)g \in A$ of smaller degree, contradiction. For uniqueness, if $qg + r = q'g + r'$, then $(q - q')g = r' - r$. If $q \ne q'$, the left has degree at least $\deg(g)$, while the right has degree $< \deg(g)$, impossible. Thus $q = q'$ and $r = r'$. □

Corollary 5.4. *Let K be a field, $f \in K[X]$, and $a \in K$. Then a is a root of f iff $X - a$ is a factor of f.*

Proof. If $X - a \mid f$, then $f(a) = 0$. Conversely, suppose $f(a) = 0$. Divide f by $X - a$: $f = q(X - a) + r$, where $r = 0$ or $\deg r < 1$, so r is constant. Substituting $X = a$ gives $0 = f(a) = r$. Hence $f = q(X - a)$. □

Corollary 5.6. *Let K be a field and $f, g \in K[X]$. Then the ideal generated by f and g equals the ideal generated by their greatest common divisor, i.e. $\langle f, g \rangle = \langle \gcd(f, g) \rangle$.*

Proof. Let $d = \gcd(f, g)$. Since $d \mid f$ and $d \mid g$, $\langle f, g \rangle \subseteq \langle d \rangle$. The Euclidean algorithm gives $d = af + bg$ for some $a, b \in K[X]$, so $d \in \langle f, g \rangle$, hence $\langle d \rangle \subseteq \langle f, g \rangle$. □

Theorem 5.8. *Let K be a field. Then every ideal of $K[X]$ is principal.*

Proof. Let $I \lhd K[X]$. If $I = \{0\} = \langle 0 \rangle$, then I is principal. Otherwise choose non-zero $g \in I$ of least degree. Let $f \in I$. By Theorem 5.1, $f = qg + r$ with $r = 0$ or $\deg(r) < \deg(g)$. Since $r = f - qg \in I$, minimality of $\deg(g)$ forces $r = 0$. Hence every $f \in I$ is a multiple of g, so $I = \langle g \rangle$. □

Lemma 6.2. *Let K be a field and $f \in K[X]$ be irreducible. Then $\langle f \rangle$ is a maximal ideal of $K[X]$.*

Proof. Let $\langle f \rangle \le I \le K[X]$. By Th[m] 5.8, $I = \langle g \rangle$. Since $f \in \langle g \rangle$, $f = qg$ for some $q \in K[X]$. As f is irreducible, q or g is invertible. If q is invertible, then $\langle g \rangle = \langle f \rangle$. If g is invertible, then $\langle g \rangle = K[X]$. Hence $\langle f \rangle$ is maximal. □

Theorem 6.3. *(Kronecker's Theorem) Let K be a field and $f \in K[X]$ be irreducible of degree n. Define $L = K[X]/\langle f \rangle$. Then: (i) L is a field and $\pi : K[X] \to L$ induces an embedding $\vartheta : K \to L$; (ii) $\alpha = X + \langle f \rangle$ is a root of f in L; (iii) L is a vector space over K of dimension n, with basis $\{1, \alpha, \ldots, \alpha^{n-1}\}$.*

Proof. (i): By Lemma 6.2, $\langle f \rangle$ is maximal, so L is a field by Corollary 4.13. Define $\vartheta : K \to L$ by $\vartheta(a) = a + \langle f \rangle$. This is a homomorphism. If $\vartheta(a) = \vartheta(b)$, then $a - b \in \langle f \rangle$. Since $\deg(f) \ge 1$, this forces $a - b = 0$, so ϑ is injective.
(ii): Let $f = b_0 + b_1 X + \cdots + b_n X^n$. For $\alpha = X + \langle f \rangle$,
$$f(\alpha) = b_0 + b_1\alpha + \cdots + b_n\alpha^n = f + \langle f \rangle = \langle f \rangle = 0_L.$$
Thus α is a root of f in L.
(iii): If $g + \langle f \rangle \in L$, divide g by f: $g = qf + r$, with $r = 0$ or $\deg(r) < n$. Writing $r = a_0 + a_1 X + \cdots + a_{n-1}X^{n-1}$, we get
$$g + \langle f \rangle = r + \langle f \rangle = a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}.$$
Thus L is spanned by $1, \alpha, \ldots, \alpha^{n-1}$. Linear independence follows since any non-zero polynomial of degree $< n$ cannot lie in $\langle f \rangle$. □

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \;\&\; n|b\}$, noting $\gcd(0, 0) = 0$ & if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \ge 2$. a is **congruent to** b **modulo** n iff $n|a - b$, written $a \equiv b \mod n$.Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \mod n\} = \{b \in \mathbb{Z} : n \mid (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$
We thus have:
$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \ldots, [n-1]_n\}$$
Note: $b \equiv a \mod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a + kn]_n$

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R, satisfying the following:
(R1) $(R, +)$ is an abelian group with identity 0.
(R2) $\times$ is associative.
(R3) $\times$ distributes over $+$ i.e.:
$$a(b + c) = ab + ac,$$
$$(a + b)c = ac + bc, \quad \forall a, b, c \in R$$
(R4) The multiplicative identity $1 \in R$ & $1 \ne 0$

Definition 1.17. Let R be a ring & $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R, 1_R.

Definition 1.21. let R be a ring. The **ring of polynomials** $R[X] = \{\sum_{i \ge 0}^n a_i X^i \text{ a finite sum} : a_i \in R\}$ with addition & multiplication defined as:
$\sum_{i\ge0}^n a_i X^i + \sum_{i\ge0}^m b_i X^i = \sum_{i\ge0}^{\max(n,m)} (a_i + b_i) X^i$
&
$(\sum_{i\ge0}^n a_i X^i)(\sum_{i\ge0}^m b_i X^i) = \sum_{i\ge0}^{n+m} (\sum_{j=0}^i a_j b_{i-j}) X^i$ (taking missing coefficients as 0)
$0 = \sum_{i\ge0} 0X^i$ & $1 = 1X^0 + \sum_{i\ge1} 0X^i$. Then the **degree** of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \ne 0$ or if no such i exist $\deg(0) = -\infty$
More genearlly, we have:
$$R[X_1, \ldots, X_n] = R[X_1, \ldots, X_{n-1}][X_n]$$
$\forall f, g \in R[X]$, $\deg(f + g) \le \max(\deg(f), \deg(g))$ & $\deg(fg) \le \deg(f) + \deg(g)$.

Definition 1.24. If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ & $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}}$;
for $n = 0$, $n \cdot a = 0 \cdot a = 0$; & for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.
The **Characteristic**, $\mathrm{char}(R)$, of a ring R is the least posititve integer n s.t. $n \cdot 1 = 0$, or if no exist $\mathrm{char}(R) = 0$.
Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.
A ring R is a domain if, $\forall r, s \in R$:
$$rs = 0 \implies r = 0 \text{ or } s = 0.$$
i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s,t being the right & left inverses, respectively.
Note: this immediately collapse, due to Le[m] 2.11, to just there exists an inverse as $s = t$.
An **invertible** element r is called a **unit**.

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r.
An element $r \in R$ is **idempotent** if $r^2 = r$.
Note: In any ring R, 0_R is nilpotent with index 1 and 0_R and 1_R are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection $\phi : R \to S$ s.t. $\forall r, r' \in R$:
$$\phi(r + r') = \phi(r) + \phi(r'),$$
$$\phi(rr') = \phi(r)\phi(r')$$
$\exists \phi : R \to S$ an isomorphism iff R & S are **isomorphic**, written $R \simeq S$.

Definition 3.6. If R & S are rings then a **homomorphism** from R to S is a map $\vartheta : R \to S$ s.t. $\forall r, r' \in R$:
$$\vartheta(r + r') = \vartheta(r) + \vartheta(r'),$$
$$\vartheta(rr') = \vartheta(r)\vartheta(r'),$$
$$\vartheta(1_R) = 1_S.$$

Definition 3.9. An **embedding** or **monomorphism** is an injective homomorphism. There is a trivial embedding along each arrow of $\mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C} \to \mathbb{C}[X]$.

Definition 3.12. If $\vartheta : R \to S$ is a ring homomorphism, then the **kernel** of ϑ is defiend as $\ker(\vartheta) = \{r \in R : \vartheta(r) = 0_S\}$.

Definition 3.16. An **automorphism** of a ring is an isomorphism from and to itself.

Definition 3.18. An **ideal** of a ring R is a subset $I \subseteq R$ s.t.: $0 \in I$; $a + b \in I, \forall a, b \in I$; & $ar, ra \in I, \forall a \in I, \forall r \in R$. We write $I \lhd R$ to mean I is an ideal of R.

Definition 3.19. If $a \in R$ then $\{r_1 a s_1 + \cdots + r_n a s_n : n \ge 1,\; r_i, s_i \in R\}$ is an ideal containing a and is the smallest ideal of R containing a. It is called the **principal ideal generated by** a, denoted $\langle a \rangle$. If R is commutative then $\langle a \rangle = \{ar : r \in R\}$. A **principal ideal** is one which can be generated by a single element.

Definition 3.21. A **right ideal** is defined as for a general ideal but with the third condition replaced by the weaker condition: $a \in I$ and $r \in R$ implies $ar \in I$. A **left ideal** is defined similarly. If $a \in R$, the **principal right ideal generated by** a is $\{ar : r \in R\}$, denoted aR.

Definition 3.22. Let $A \subseteq R$ and R be a commutative ring, then $\{a_1 r_1 + \cdots + a_n r_n : n \ge 1, a_i \in A, r_i \in R\}$ is the **ideal generated by** A.

Definition 3.26. We can also define a **product of ideals**: if $I, J \lhd R$, then their product is the ideal generated by $\{ab : a \in I, b \in J\}$. This set will not in general be closed under addition, hence we say "generated by". Powers of an ideal are defined inductively by $I^{n+1} = I^n I$.

Definition 4.1. Let R be a ring and let I be a proper ideal. Let R/I denote the set of cosets of I in the additive group $(R, +)$:
$$R/I = \{r + I : r \in R\}.$$
Define operations $+$ and $\times$ on R/I by
$$(r + I) + (s + I) = (r + s) + I, \quad (r + I)(s + I) = rs + I.$$

Definition 4.1. The ring R/I is called the **factor ring** (or **quotient ring**) of R by I.

Exercise. Write out the elements of the factor ring $\mathbb{Z}_6/\langle [3]_6 \rangle$. What is this factor ring isomorphic to?

Definition 4.4. The map $\pi : R \to R/I$, $\pi(r) = r + I$, is called the **canonical projection**, or **canonical surjection**.
Notation: if $I \subseteq J$ are ideals then usually write $I \le J$ (or $J \ge I$) instead of subset notation, with the same meaning.
The notation J/I is also used instead of πJ for the image of J in R/I.

Definition 4.12. An ideal I of a ring R is **maximal** if it is proper and for any ideal J with $I \le J \le R$, either $J = I$ or $J = R$.

Definition 4.14. A proper ideal I of a commutative ring R is **prime** if whenever $r, s \in R$ and $rs \in I$, then $r \in I$ or $s \in I$.

Definition 5.3. An element $a \in K$ is a **root** (or **zero**) of $f \in K[X]$ iff $f(a) = 0$.

Definition 5.5. The **greatest common divisor** of polynomials f, g is a polynomial d such that $d \mid f$ and $d \mid g$, and if $h \mid f$ and $h \mid g$, then $h \mid d$. This determines d up to a non-zero scalar; for uniqueness take d **monic**.

Division Algorithm for Polynomials. For $f, g \in K[X]$, $g \ne 0$, repeatedly apply Th[m] 5.1:
$$\begin{aligned} f &= q_1 g + r_1, & \deg(r_1) &< \deg(g), \\ g &= q_2 r_1 + r_2, & \deg(r_2) &< \deg(r_1), \\ r_1 &= q_3 r_2 + r_3, & \deg(r_3) &< \deg(r_2), \\ &\;\;\vdots \\ r_{n-1} &= q_{n+1} r_n. \end{aligned}$$
Since degrees strictly decrease, some $r_{n+1} = 0$. Then
$$\gcd(f, g) = \gcd(g, r_1) = \cdots = \gcd(r_{n-1}, r_n) = r_n$$
up to multiplication by a non-zero scalar; take the monic version.
Tracking back expresses $\gcd(f, g) = kf + lg$ for some $k, l \in K[X]$.

Factorisation. Here R is commutative.

Definition 5.9. (i) $r \in R$ is **irreducible** if r is not invertible and whenever $r = st$, either s or t is invertible. (ii) $r, s \in R$ are **associated** if $s = ur$ for some invertible $u \in R$.

Definition 5.11. A commutative domain R is a **Unique Factorisation Domain (UFD)** if every non-zero, non-invertible element factors uniquely into irreducibles, up to rearrangement and associated factors.

Definition 5.14. A **Principal Ideal Domain (PID)** is a commutative domain in which every ideal is principal.
To produce a root for a polynomial p, it is enough to produce a root of one irreducible factor of p. Hence focus on irreducible polynomials.

Examples

Example. With $n \ge 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{\mathbf{0}\}$ but this isn't a division ring hence even relaxing commutative in Pro[p] 3.22 doesn't yeild the same pro[p] with field $\to$ division ring.

Example. For X a set $\mathcal{P}(X)$ with
$$A + B = (A \cup B) \setminus (A \cap B),$$
$$AB = A \cap B$$
& $0 = \varnothing$, $-A = A$, $A, B \in \mathcal{P}(X)$.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ & $ij = k$, $jk = i$, & $ki = j$, this is non-commutative as $-ij = -k = jjk = ji$ so $ij \ne ji$ and this is a division ring as for $z = a + bi + cj + dk$ $z^{-1} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}$, taking $z \ne 0$.

Example. For 4 element rings $\mathbb{Z}_4 \not\simeq \mathbb{Z}_2 \times \mathbb{Z}_2$, since given $[2]_4 \ne 0$ and $[2]_4^2 = [0]_4$, $[2]_4$ is nilpotent, where as there are no non-zero nilpotent elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$, hence the result by Le[m] 3.3.

Example. The map $\mathbb{Z} \to \mathbb{Z}_n$ by $m \mapsto [m]_n$ is a homomorphism. Also, $\vartheta : \mathbb{Z}_{kn} \to \mathbb{Z}_n$ by $\vartheta([m]_{kn}) = [m]_n$ is a homomorphism, well definedness must be shown, and is true as; let $[m]_{kn} = [m']_{kn} \in \mathbb{Z}_{kn}$ then $kn \mid m - m'$ so $n|m - m'$ thus $[m]_n = [m']_n$.

Example. With R a ring, the map $R[X] \to R[X]$ by $p(X) \mapsto p(q(X))$ for some $q(X) \in R[X]$.

Example. $\mathbb{Z}[i] \to \mathbb{Z}[i]$ by $a + bi \mapsto a - bi$ is an automorphism.

Example. Examples 3.20. (i) In every ring $\langle 0 \rangle = \{0\}$ is the smallest ideal, the **trivial ideal**. (ii) In every ring $\langle 1 \rangle = R$ is an ideal and every other ideal $\ne R$ is a **proper ideal**. (iii) If $n \in \mathbb{Z}$ then $\langle n \rangle = \{nk : k \in \mathbb{Z}\}$. (iv) If $p \in R[X]$ then $\langle p \rangle$ is the set of polynomials with p as a factor.

Example. Let $R = M_2(\mathbb{Z})$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ then by simple multiplication the principle left and right ideals are $\{\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ and $\{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ whilst $\langle A \rangle = M_2(\mathbb{Z})$ as $I_2 = A + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \langle A \rangle$
Note: If $1 \in I$ with $I \lhd R$ the $\forall r \in R$ $r = r1 \in I$, so $I = R$.

Example. Working in $\mathbb{Z}[X]$ the ideal $I = \langle 2, X \rangle := \langle \{2, X\} \rangle$ is not a principal ideal. Suppose $I = \langle p \rangle$. Since $2 \in I$, $2 = pq$ for some $q \in \mathbb{Z}[X]$. As $\mathbb{Z}$ is a domain, $0 = \deg(2) = \deg(pq) = \deg(p) + \deg(q)$, so $p, q \in \mathbb{Z}$ and $p = \pm 1$ or $p = \pm 2$. If $p = \pm 1$, then $1 = 2r_1 + Xr_2$ for some $r_1, r_2 \in \mathbb{Z}[X]$, impossible since the RHS has even constant term. Hence $p = \pm 2$. But $X \in I$, so $X = 2r_3$ for some $r_3 \in \mathbb{Z}[X]$, impossible since $2r_3$ has even coefficients. Thus I is not principal.

Example. $\mathbb{Z}/\langle n \rangle = \mathbb{Z}_n$.

Example. 4.6. Take $R = \mathbb{Z}$ and $I = \langle 6 \rangle = 6\mathbb{Z}$, so the canonical projection is $\pi : \mathbb{Z} \to \mathbb{Z}_6$. Let $\vartheta : \mathbb{Z} \to \mathbb{Z}/\langle 2 \rangle = \mathbb{Z}_2$ be the canonical projection. Then $\ker(\vartheta) = \langle 2 \rangle \supseteq \langle 6 \rangle$, so by Le[m] 4.4 there is a unique homomorphism ϑ' with $\vartheta' \circ \pi = \vartheta$. This is the map $\vartheta' : \mathbb{Z}_6 \to \mathbb{Z}_2$ already seen in Examples 3.8.

Example. 4.7. Let $R = \mathbb{R}[X]$ and $I = \langle X^2 + 1 \rangle$. Let $\pi : \mathbb{R}[X] \to \mathbb{R}[X]/\langle X^2 + 1 \rangle$ be the canonical surjection. Let $\vartheta : \mathbb{R}[X] \to \mathbb{C}$ be evaluation at i, $\vartheta(p(X)) = p(i)$. Then $p \in \ker(\vartheta)$ iff $X^2 + 1 \mid p$, so $\ker(\vartheta) = \langle X^2 + 1 \rangle$. By Le[m] 4.4, there is a unique map $\vartheta' : \mathbb{R}[X]/\langle X^2 + 1 \rangle \to \mathbb{C}$ through which ϑ factors. It is injective and surjective, hence an isomorphism. Thus $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Example. 4.9. The ideals of $\mathbb{Z}_6 = \mathbb{Z}/\langle 6 \rangle$ are in one-to-one correspondence with the ideals of $\mathbb{Z}$ containing $\langle 6 \rangle$. There are 4 of each: $\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle$ correspond to
$$\begin{aligned} \pi\langle 1 \rangle &= \mathbb{Z}_6 = \{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\}, \\ \pi\langle 2 \rangle &= \langle [2]_6 \rangle = \{[0]_6, [2]_6, [4]_6\}, \\ \pi\langle 3 \rangle &= \langle [3]_6 \rangle = \{[0]_6, [3]_6\}, \\ \pi\langle 6 \rangle &= \langle [0]_6 \rangle = \{[0]_6\}. \end{aligned}$$

Example. 4.11. Since $\langle 6 \rangle \le \langle 2 \rangle$, Th[m] 4.10 gives
$$(\mathbb{Z}/\langle 6 \rangle)/(\langle 2 \rangle/\langle 6 \rangle) \simeq \mathbb{Z}/\langle 2 \rangle.$$

Example. 4.15. All ideals of $\mathbb{Z}$ are principal. The prime ideals are precisely the ideals generated by a prime number or zero.

Example. 5.2. In $\mathbb{Z}_5[X]$,
$$X^4 + 2X^2 + X + 4 = (X^2 + 2X + 3)(X^2 + 3X + 3) + X.$$

Example. Examples 5.10. (i) In $\mathbb{Z}$, irreducibles are $\pm p$ for p prime; r, s are associated iff $r = \pm s$. (ii) In $K[X]$, K a field, irreducibles are non-constant polynomials with no factor g satisfying $1 \le \deg(g) < \deg(f)$.

Example. 5.12. $\mathbb{Z}$ is a UFD. For example, $-24 = (-2) \cdot 2 \cdot (-2) \cdot (-3)$, unique up to signs and ordering.

Example. 5.13. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD since $6 = 3 \cdot 2 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, with irreducible factors not matching up to associates.

Example. Examples 5.15. $\mathbb{Z}$ is a PID; if K is a field, $K[X]$ is a PID by Th[m] 5.8. Every PID is a UFD. If R is a UFD then so is $R[X]$, hence $R[X_1, \ldots, X_n]$.

Example. $\mathbb{Z}[X]$ is a UFD but not a PID: it is a UFD since $\mathbb{Z}$ is, but $\langle 2, X \rangle$ is not principal.

Example. Examples 5.16. (i) $X^3 + X^2 + 1$ is irreducible in $\mathbb{Z}_2[X]$: a reducible cubic has a linear factor, but $0, 1$ are not roots. (ii) $X^4 + 2X^3 + X + 1$ is irreducible in $\mathbb{Z}_3[X]$: no roots, and comparing coefficients in two possible quadratics gives no factorisation.

Example. 6.1. Take $p = X^2 - 2 \in \mathbb{Q}[X]$. An extension field containing a root is $\mathbb{R}$, or more efficiently $\mathbb{Q}[\sqrt{2}]$.

Example. 6.4. (i) If $K = \mathbb{Q}$ and $f = X^2 - 2$, then $\mathbb{Q}[X]/\langle X^2 - 2 \rangle \simeq \mathbb{Q}[\sqrt{2}]$. (ii) If $K = \mathbb{R}$ and $f = X^2 + 1$, every element of $\mathbb{R}[X]/\langle X^2 + 1 \rangle$ has form $a_0 + a_1 i$, using i for $X + \langle f \rangle$, so $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Example. 6.5. If $f = X(X - 1)$, then $K[X]/\langle X(X - 1) \rangle$ is not a field. With $\alpha = X + \langle f \rangle$, neither α nor $\alpha - 1$ is zero, but $\alpha(\alpha - 1) = 0$. So the quotient is not even a domain.

Example. 6.6. For $X^3 + X + 1 \in \mathbb{Z}_2[X]$, there are no roots in $\mathbb{Z}_2$, so it is irreducible. Let $\alpha = X + \langle f \rangle$ and $L = \mathbb{Z}_2[X]/\langle f \rangle$. By Kronecker, L is a field with 2^3 elements:
$$\{0, 1, \alpha, \alpha^2, 1 + \alpha, 1 + \alpha^2, \alpha + \alpha^2, 1 + \alpha + \alpha^2\}.$$
In L, $\alpha^3 + \alpha + 1 = 0$, so $\alpha^3 = \alpha + 1$, which determines the addition and multiplication tables.

